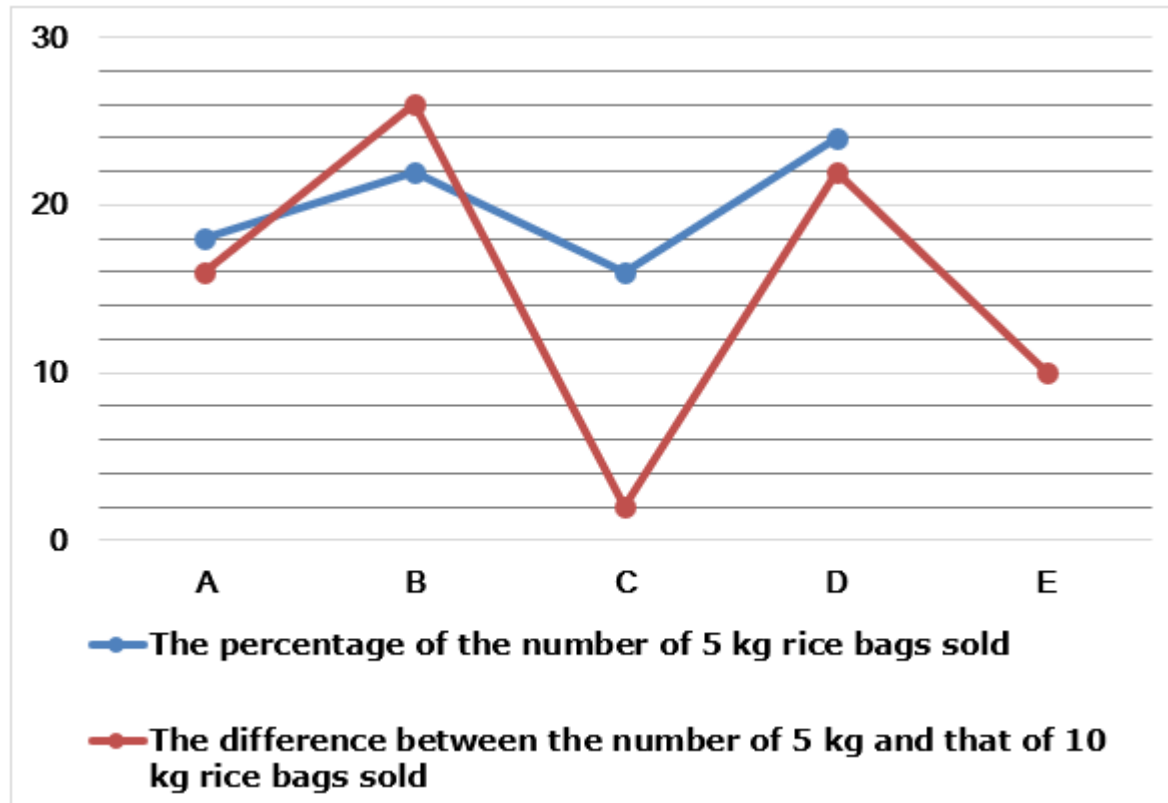


1. Questions

Study the following information carefully and answer the questions given below:

The line graph below shows the percentage of the number of 5 kg rice bags sold out of the total number of 5 kg rice bags sold and the difference between the number of 5 kg and that of 10 kg rice bags sold in five different shops i.e. A, B, C, D and E.



Notes:

i). The total number of 5 kg rice bags sold in all five shops together is more than 560.

ii). The total number of 5 kg and 10 kg rice bags sold in shop E is 230.

If the total number of 10 kg rice bags in shop B is 66 more than that of shop A. The total number of rice bags in shop B is what percentage of that of shop A?

Note: The total number of rice bags in shop B is < 300.

- 125%
- 150%
- 145%
- 175%
- 120%

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2. Questions

What is the possible value for the average number of rice bags sold in shop C?

I). 93

II). 95

III). 96

IV). 97

- a. Both I and II
- b. Both II and IV
- c. Both I and III
- d. Both III and IV
- e. Both I and IV

3. Questions

If the number of 10 kg rice bags sold in shops B and E is Rs. 450 per bag and Rs. 500 per bag, respectively. If the total revenue of 10 kg rice bags sold in shops B and E together is Rs. 126100. Find the difference between the total number of rice bags sold in shop B and the number of 5 kg rice bags sold in shop E.

- a. 170
- b. 210
- c. 150
- d. 140
- e. 190

4. Questions

What is the ratio of the maximum possible number of 10 kg rice bags sold in shop D to the sum of the 5 kg rice bags sold in shops C and E together?

- a. 81: 103
- b. 83: 105
- c. 87: 108
- d. 83: 108
- e. 81: 101

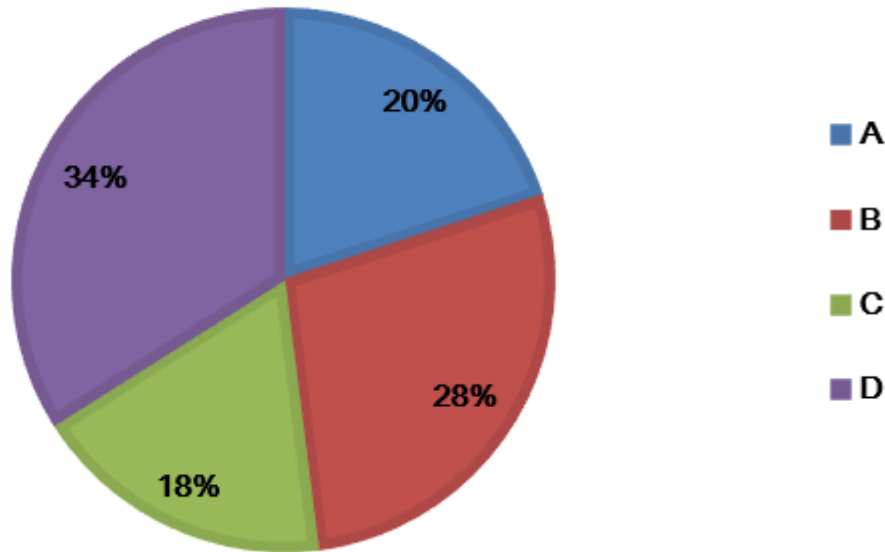
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5. Questions

Study the following information carefully and answer the questions given below:

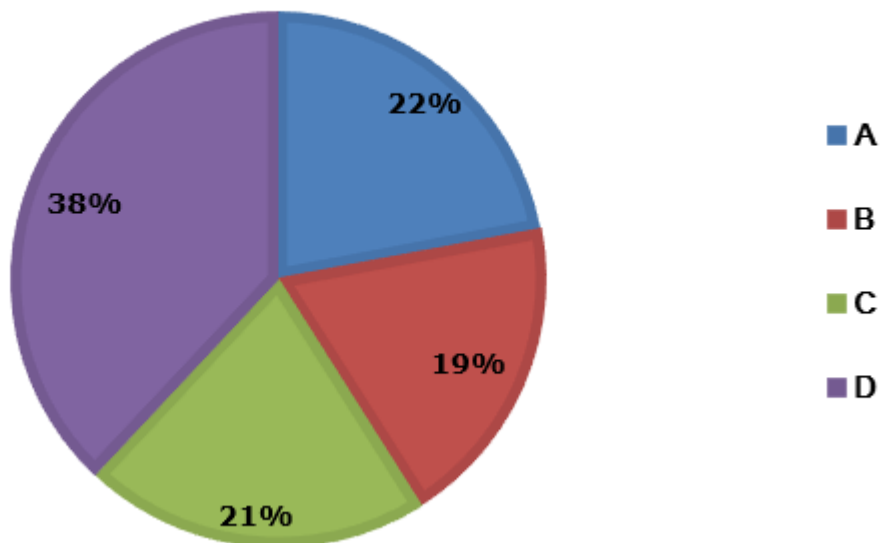
The pie chart below shows the percentage of the weight of gold (in grams) out of the total weight of gold combined in four different shops i.e. A, B, C and D.

The weight of gold (grams)



The pie chart below shows the percentage of the weight of silver sold (in grams) out of the total weight of silver sold combined in four different shops on Monday i.e. A, B, C and D.

The weight of silver sold (in grams)



Notes:

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i). The percentage of the weight of gold (in grams) in each shop out of the total weight of gold combined in four different shops on Monday is the same as the percentage of the weight of silver (in grams) in each shop out of the total weight of silver combined in four different shops, respectively.

ii). The weight of silver unsold in shops A and B is 240 grams and 690 grams, respectively.

iii). The total weight of gold in shop D is 74 grams more than that of the total weight of silver in shop C.

If the weight of gold sold in shop B is 227.5% of the weight of silver unsold in shop A. What is the total weight of unsold gold and silver together in shop B?

- a. 872 grams
- b. 888 grams
- c. 772 grams
- d. 722 grams
- e. 812 grams

6. Questions

If the ratio of the price per gram of gold and that in silver in shop C is 80:1, respectively, and the total selling amount of gold and silver together in shop C is Rs. 2303100. The weight of gold unsold in shop C is 75% of that in shop A. Find the weight of gold sold in shop A.

Note: If the selling amount of silver in shop C is Rs. 56700.

- a. 272 grams
- b. 378 grams
- c. 212 grams
- d. 222 grams
- e. 312 grams

7. Questions

If the ratio of the weight of gold unsold in shop B to that in shop D is 91:136, respectively, and the sum of the weight of gold sold in shops B and D together is 1158 grams. Find the ratio of the difference between the weight of gold unsold in shop B to that in shop D to the difference between the weight of silver unsold in shop B to that in shop D.

- a. 7: 10
- b. 1: 5
- c. 3: 10
- d. 7: 15
- e. 3: 8

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8. Questions

If the sum of the weight of gold sold in all four shops together is 59.4% of that of silver in all four shops together. The sum of the weight of gold unsold in all four shops together is what percentage of the sum of the weight of silver in all four shops together?

- a. 20%
- b. (818/45)%
- c. 22.5%

d. $(411/36)\%$

e. $(212/9)\%$

9. Questions

Study the following information carefully and answer the questions given below:

The table below shows the number of new students who appeared for the exam, the percentage of students who passed out of the total number of students who appeared for the exam, and the cumulative number of students who failed the exam across four different attempts.

Attempts	The number of new students who appeared for the exam	The percentage of students who passed out of the total number of students who appeared for the exam	The cumulative number of students who failed the exam
1	280	$X\%$	-
2	274	$U\%$	89
3	182	80%	-
4	-	$'X + 12'\%$	156

Notes:

i). The total number of students who appeared for the exam = The number of new students who appeared for the exam + the number of students who failed the exam for the previous attempt.

ii). The total number of students who appeared for the exam in attempt 1 is the number of new students who appeared for the exam.

iii). The sum of ' X ' and ' U ' is 170 and the value of ' $U - X$ ' is 10.

If the ratio of the number of male to female students who failed the exam in attempt 2 is 6:5, respectively, and the number of male students who passed the exam in attempt 2 is $2X$. Find the number of female students who appeared the exam in attempt 2.

a. $X + 42$

b. $2U - 18$

c. $X + 52$

d. $2X - 8$

e. $U - X + 87$

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10. Questions

In attempt 3, 85.71% of the new students who took the exam passed. The remaining students who passed were those who had failed in the previous attempt. Find the difference between the number of students who failed the exam from the number of new students who appeared for the exam, and the number of students who failed again from the previous attempt.

- a. 8
- b. 16
- c. 9
- d. 18
- e. 7

11. Questions

The difference between the number of students who passed the exam in attempts 2 and 4 is what percentage of the total number of students who appeared for the exam in attempt 4?

- a. 18%
- b. 14%
- c. 5%
- d. 21%
- e. 7%

12. Questions

If the number of new students who appeared for the exam in attempt 5 is 140% of U and the number of students who passed the exam in attempt 5 is $[(X/2) + 10]\%$ of that of attempt 4. Find the number of students who failed the exam in attempt 5.

- a. 12
- b. 24
- c. 18
- d. 15
- e. 27

13. Questions

A and B invested Rs. $(X-10000)$ and Rs. $(X + 2000)$ in a $R\%$ compound half yearly for 1 year and $1.5R\%$ compounded annually for 2 years, respectively. If the ratio of the investments of A and B is 5:8, and the interest earned by A is Rs. 2050, then find the difference between their interests.

- a. Rs. 8920
- b. Rs. 7340
- c. Rs. 7770
- d. Rs. 8270
- e. Rs. 8140

14. Questions

The cost price and selling price of the article are Rs. a and Rs. $1.8a$, respectively, and the marked-up percentage of the article is 100%. If the cost price is increased by Rs. 100 and the selling price is decreased by Rs. 200, the marked price of the article remains the same. Find the difference between the new profit amount and the new discount amount of the article. ($1000 \geq a < 1300$, and 'a' is a perfect cube)

- a. Rs. 100
- b. Rs. 200
- c. Rs. 300
- d. Rs. 400
- e. Rs. 500

15. Questions

A and B together to complete the work in $P/15$ days, C and B together to complete the work in $P/8$ days, and A alone to complete the whole work in $P/10$ days. If C alone can complete the work in 50 days, then find the value of $[(P/10)^2 - 1.2P]$.

- a. $P - 20$
- b. $P - 105$
- c. $P + 20$
- d. $P + 35$
- e. $P + 15$

16. Questions

The sum of the radius and height of the cone is 49 cm, and the height of the cone is 10 cm more than the radius of the sphere, whose volume is $7776\pi \text{ cm}^3$. Find the lateral surface area of the cone.

- a. 1848 cm^2
- b. 1188 cm^2
- c. 2310 cm^2
- d. 1584 cm^2
- e. 3080 cm^2

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17. Questions

The following questions have two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give answer as,

The time taken by train Q to cross the pole is 2 times that of train P to cross the pole. The speed of train Q is 70% of the speed of train P.

Quantity I: Find the length of the train P. If the difference between the lengths of both trains P and Q is 192 m.

Quantity II: The time taken by train P to cross the 240 m platform is 1.5 times the time taken by train P to cross the pole. Find the length of the bridge which is 172 m less than that of train Q.

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

18. Questions

The smaller root of the equation ' $4p^2 + 19p + 56 * 24 = 2p^2 + 66p + 1159$ ' is y, and $y/(97 * 8 - 257 * 3) = 27/(z + 12)$.

Quantity I: 100y

Quantity II: If the value of 'x' is 80% of the value of 'z'. Find the value of ' $16.7x + 20z$ '.

- a. Quantity I > Quantity II
- b. Quantity II $\geq$ Quantity I
- c. Quantity II > Quantity I
- d. Quantity I $\geq$ Quantity II
- e. Quantity I = Quantity II or Relation cannot be established

19. Questions

There are three persons in a family. The present age of P is 6 years more than that of A, and the present age of P is 3 times the present age of Y. At the time of Y's birth, A's age is 80% of P's age.

Quantity I: Find the average age of the present age of P and the present age of A.

Quantity II: If the ratio of the present age of S to the present age of Y is 13:3, respectively. Find the age of S 24 years ago.

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

20. Questions

Prabhu invested Rs. X and Banu invested Rs. Y in the business. After five months, Prabhu increased his investment by 20%, and after one more month, Banu decreased his investment by 25%. The total profit earned in the business in one year is Rs. 15600, and the profit share of Prabhu is Rs. 8040.

Quantity I: If the sum of the values of 'X' and 'Y' is Rs. 11000. Find the value of 'X'.

Quantity II: If Rs. Y is invested in scheme at 8% simple interest per annum and interest received after 3 years is Rs. 1440. Find the value of 'X'.

- a. Quantity I > Quantity II
- b. Quantity I $\geq$ Quantity II
- c. Quantity II > Quantity I
- d. Quantity II $\geq$ Quantity I
- e. Quantity I = Quantity II or Relation cannot be established

Explanations:

1. Questions

The total number of 5 kg and 10 kg rice bags sold in shop E is 230.

Let assume the number of 5 kg and 10 kg rice bags sold in shop E are 'X' and 'Y' respectively

$$X + Y = 230 \text{ ----- (1)}$$

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop E

The possible cases are

$$X - Y = 10 \text{ (or) } Y - X = 10$$

Case (i):

$$Y - X = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2Y = 240$$

$$Y = 120$$

And

$$120 - X = 10$$

$$X = 110$$

So, the number of 5 kg rice bags sold in shop E = 110

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The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(110/20) * 100 = 550$

$550 < 560$

This is invalid.

Case (ii):

$$X - Y = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2X = 240$$

$$X = 120$$

So, the number of 5 kg rice bags sold in shop E = 120

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(120/20) * 100 = 600$

$600 > 560$

This is valid.

Let the number of 5 kg rice bags sold in shop A = 18% of $600 = 108$

The number of 5 kg rice bags sold in shop B = 22% of $600 = 132$

The number of 5 kg rice bags sold in shop C = 16% of $600 = 96$

The number of 5 kg rice bags sold in shop D = 24% of $600 = 144$

Let the table

Shop	The number of 5 kg rice bags sold	The number of 10 kg rice bags sold	The total number of rice bags sold
A	108	-	-
B	132	-	-
C	96	-	-
D	144	-	-
E	120	110	230

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Answer: C

Let,

The number of 5 kg rice bags sold in shop A = 108

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop A = 16

If (5 kg rice bags < 10 kg rice bags)

The number of 10 kg rice bags sold in shop A = $108 + 16 = 124$

The number of 10 kg rice bags sold in shop B = $124 + 66 = 190$

The total number of rice bags in shop B = $190 + 132 = 322$

This is not satisfied the total number of rice bags in shop B is < 300.

If (5 kg rice bags > 10 kg rice bags)

The number of 10 kg rice bags sold in shop A = $108 - 16 = 92$

The number of 10 kg rice bags sold in shop B = $92 + 66 = 158$

The total number of rice bags in shop B = $158 + 132 = 290$

This is satisfied the total number of rice bags in shop B is < 300.

So, The total number of rice bags in shop A = $92 + 108 = 200$

Thus, percentage = $(290/200) * 100 = 145\%$

Hence, option C is correct.

2. Questions

The total number of 5 kg and 10 kg rice bags sold in shop E is 230.

Let assume the number of 5 kg and 10 kg rice bags sold in shop E are 'X' and 'Y' respectively

$$X + Y = 230 \text{ ----- (1)}$$

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop E

The possible cases are

$$X - Y = 10 \text{ (or) } Y - X = 10$$

Case (i):

$$Y - X = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2Y = 240$$

$$Y = 120$$

And

$$120 - X = 10$$

$$X = 110$$

So, the number of 5 kg rice bags sold in shop E = 110

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

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The total number 5 kg rice bags sold in all five shops together = $(110/20) * 100 = 550$

$$550 < 560$$

This is invalid.

Case (ii):

$$X - Y = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2X = 240$$

$$X = 120$$

So, the number of 5 kg rice bags sold in shop E = 120

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(120/20) * 100 = 600$

$$600 > 560$$

This is valid.

Let the number of 5 kg rice bags sold in shop A = 18% of $600 = 108$

The number of 5 kg rice bags sold in shop B = 22% of $600 = 132$

The number of 5 kg rice bags sold in shop C = 16% of $600 = 96$

The number of 5 kg rice bags sold in shop D = 24% of $600 = 144$

Let the table

Shop	The number of 5 kg rice bags sold	The number of 10 kg rice bags sold	The total number of rice bags sold
A	108	-	-
B	132	-	-
C	96	-	-
D	144	-	-
E	120	110	230

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Answer: B

The number of 5 kg rice bags sold in shop C = 96

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop C = 2

If (5 kg rice bags < 10 kg rice bags)

The number of 10 kg rice bags sold in shop C = $96 + 2 = 98$

The average number of rice bags in shop C = $(96 + 98)/2 = 97$

If (5 kg rice bags > 10 kg rice bags)

The number of 10 kg rice bags sold in shop C = $96 - 2 = 94$

The average number of rice bags in shop C = $(96 + 94)/2 = 95$

So, the possible values are 95 and 97.

Hence, option B is correct.

3. Questions

The total number of 5 kg and 10 kg rice bags sold in shop E is 230.

Let assume the number of 5 kg and 10 kg rice bags sold in shop E are 'X' and 'Y' respectively

$$X + Y = 230 \text{ ----- (1)}$$

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop E

The possible cases are

$$X - Y = 10 \text{ (or) } Y - X = 10$$

Case (i):

$$Y - X = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2Y = 240$$

$$Y = 120$$

And

$$120 - X = 10$$

$$X = 110$$

So, the number of 5 kg rice bags sold in shop E = 110

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(110/20) * 100 = 550$

$$550 < 560$$

This is invalid.

Case (ii):

$$X - Y = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2X = 240$$

$$X = 120$$

So, the number of 5 kg rice bags sold in shop E = 120

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(120/20) * 100 = 600$

$$600 > 560$$

This is valid.

Let the number of 5 kg rice bags sold in shop A = 18% of $600 = 108$

The number of 5 kg rice bags sold in shop B = 22% of $600 = 132$

The number of 5 kg rice bags sold in shop C = 16% of $600 = 96$

The number of 5 kg rice bags sold in shop D = 24% of $600 = 144$

Let the table

Shop	The number of 5 kg rice bags sold	The number of 10 kg rice bags sold	The total number of rice bags sold
A	108	-	-
B	132	-	-
C	96	-	-
D	144	-	-
E	120	110	230

Answer: A

The number of 10 kg rice bags sold shop B = B

Let,

$$B * 450 + 110 * 500 = 126100$$

$$50 * (9B + 1100) = 126100$$

$$9B + 1100 = 2522$$

$$9B = 1422$$

$$B = 158$$

The total number of rice bags sold in shop B = $158 + 132 = 290$

Thus, difference = $290 - 120 = 170$

Hence, option A is correct.

4. Questions

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The total number of 5 kg and 10 kg rice bags sold in shop E is 230.

Let assume the number of 5 kg and 10 kg rice bags sold in shop E are 'X' and 'Y' respectively

$$X + Y = 230 \text{ ----- (1)}$$

The difference between the number of 5 kg and that of 10 kg rice bags sold in shop E

The possible cases are

$$X - Y = 10 \text{ (or) } Y - X = 10$$

Case (i):

$$Y - X = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2Y = 240$$

$$Y = 120$$

And

$$120 - X = 10$$

$$X = 110$$

So, the number of 5 kg rice bags sold in shop E = 110

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(110/20) * 100 = 550$

$$550 < 560$$

This is invalid.

Case (ii):

$$X - Y = 10$$

$$\text{WKT, } X + Y = 230$$

Solve the above two equations

$$2X = 240$$

$$X = 120$$

So, the number of 5 kg rice bags sold in shop E = 120

The percentage of the number of rice bags sold in shop E = $100\% - (18 + 22 + 16 + 24)\% = 20\%$

The total number 5 kg rice bags sold in all five shops together = $(120/20) * 100 = 600$

$$600 > 560$$

This is valid.

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Let the number of 5 kg rice bags sold in shop A = 18% of 600 = 108

The number of 5 kg rice bags sold in shop B = 22% of 600 = 132

The number of 5 kg rice bags sold in shop C = 16% of 600 = 96

The number of 5 kg rice bags sold in shop D = 24% of 600 = 144

Let the table

Shop	The number of 5 kg rice bags sold	The number of 10 kg rice bags sold	The total number of rice bags sold
A	108	-	-
B	132	-	-
C	96	-	-
D	144	-	-
E	120	110	230

Answer: D

The maximum possible number of 10 kg rice bags sold in shop D (5 kg rice bags < 10 kg rice bags)

The maximum possible number of 10 kg rice bags sold in shop D = 144 + 22 = 166

The sum of number 5 kg rice bags sold in shops C and E together = 96 + 120 = 216

Thus, ratio = 166: 216 = 83: 108

Hence, option D is correct.

5. Questions

From pie chart 2,

The total weight of silver sold combined in all four shops together = 100x

The weight of silver sold in shop A = 22% of 100x = 22x grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = 22x + 240 grams

The weight of silver sold in shop B = 19% of 100x = 19x grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = 19x + 690 grams

The percentage of the total weight of silver in each shop is same as that of gold in each shop respectively.

The percentage of the total weight of gold in shop A = 20%

The percentage of the total weight of gold in shop B = 28%

Let,

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$$(22x + 240)/(19x + 690) = 20/28$$

$$(22x + 240) * 7 = (19x + 690) * 5$$

$$154x + 1680 = 95x + 3450$$

$$59x = 1770$$

$$x = 30$$

Let,

The weight of silver sold in shop A = $22x = 22 * 30 = 660$ grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $660 + 240 = 900$ grams

The weight of silver sold in shop B = $19x = 19 * 30 = 570$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $570 + 690 = 1260$ grams

So,

The percentage of the total weight of silver in shop C = 18%

The total weight of silver in shop C = $(900/20) * 18 = 810$ grams

The weight of silver sold in shop C = $(660/22) * 21 = 630$ grams

The weight of silver unsold in shop C = $810 - 630 = 180$ grams

The percentage of the total weight of silver in shop D = 34%

The total weight of silver in shop D = $(900/20) * 34 = 1530$ grams

The weight of silver sold in shop D = $(660/22) * 38 = 1140$ grams

The weight of silver unsold in shop D = $1530 - 1140 = 390$ grams

Let,

The total weight of gold in shop D is 74 grams more than that of the total weight of silver in shop C.

The total weight of gold in shop D = $810 + 74 = 884$ grams

Then,

The total weight of gold in shop A = $(884/34) * 20 = 520$ grams

The total weight of gold in shop B = $(884/34) * 28 = 728$ grams

The total weight of gold in shop C = $(884/34) * 18 = 468$ grams

Let table,

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Shop	Gold (in grams)			Silver (in grams)		
	Total	Sold	Unsold	Total	Sold	Unsold
A	520	-	-	900	660	240
B	728	-	-	1260	570	690
C	468	-	-	810	630	180
D	884	-	-	1530	1140	390

Answer: A

Let,

The weight of gold sold in shop B = 227.5% of 240 = 546 grams

The weight of gold unsold in shop B = 728 – 546 = 182 grams

The total weight of unsold gold and silver together in shop B = 182 + 690 = 872 grams

Hence, option A is correct.

6. Questions

From pie chart 2,

The total weight of silver sold combined in all four shops together = 100x

The weight of silver sold in shop A = 22% of 100x = 22x grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = 22x + 240 grams

The weight of silver sold in shop B = 19% of 100x = 19x grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = 19x + 690 grams

The percentage of the total weight of silver in each shop is same as that of gold in each shop respectively.

The percentage of the total weight of gold in shop A = 20%

The percentage of the total weight of gold in shop B = 28%

Let,

$$(22x + 240)/(19x + 690) = 20/28$$

$$(22x + 240) * 7 = (19x + 690) * 5$$

$$154x + 1680 = 95x + 3450$$

$$59x = 1770$$

$$x = 30$$

Let,

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The weight of silver sold in shop A = $22x = 22 * 30 = 660$ grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $660 + 240 = 900$ grams

The weight of silver sold in shop B = $19x = 19 * 30 = 570$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $570 + 690 = 1260$ grams

So,

The percentage of the total weight of silver in shop C = 18%

The total weight of silver in shop C = $(900/20) * 18 = 810$ grams

The weight of silver sold in shop C = $(660/22) * 21 = 630$ grams

The weight of silver unsold in shop C = $810 - 630 = 180$ grams

The percentage of the total weight of silver in shop D = 34%

The total weight of silver in shop D = $(900/20) * 34 = 1530$ grams

The weight of silver sold in shop D = $(660/22) * 38 = 1140$ grams

The weight of silver unsold in shop D = $1530 - 1140 = 390$ grams

Let,

The total weight of gold in shop D is 74 grams more than that of the total weight of silver in shop C.

The total weight of gold in shop D = $810 + 74 = 884$ grams

Then,

The total weight of gold in shop A = $(884/34) * 20 = 520$ grams

The total weight of gold in shop B = $(884/34) * 28 = 728$ grams

The total weight of gold in shop C = $(884/34) * 18 = 468$ grams

Let table,

Shop	Gold (in grams)			Silver (in grams)		
	Total	Sold	Unsold	Total	Sold	Unsold
A	520	-	-	900	660	240
B	728	-	-	1260	570	690
C	468	-	-	810	630	180
D	884	-	-	1530	1140	390

Answer: E

The ratio of the price per gram of gold and that in silver in shop C on Monday = 80:1 = $80x : x$

The selling amount of silver in shop C = Rs. 56700

The price per gram of silver in shop C = $56700/630$ = Rs. 90 per gram

The price per gram of gold in shop C = $(90/1) * 80$ = Rs. 7200 per gram

The selling amount of gold in shop C = $2303100 - 56700$ = Rs. 2246400

The weight of gold sold in shop C = $2246400/7200$ = 312

The weight of gold unsold in shop C = $468 - 312$ = 156 grams

The weight of gold unsold in shop A = $(156/75) * 100$ = 208 grams

The weight of gold sold in shop A = $520 - 208$ = 312 grams

Hence, option E is correct.

7. Questions

From pie chart 2,

The total weight of silver sold combined in all four shops together = $100x$

The weight of silver sold in shop A = 22% of $100x$ = $22x$ grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $22x + 240$ grams

The weight of silver sold in shop B = 19% of $100x$ = $19x$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $19x + 690$ grams

The percentage of the total weight of silver in each shop is same as that of gold in each shop respectively.

The percentage of the total weight of gold in shop A = 20%

The percentage of the total weight of gold in shop B = 28%

Let,

$$(22x + 240)/(19x + 690) = 20/28$$

$$(22x + 240) * 7 = (19x + 690) * 5$$

$$154x + 1680 = 95x + 3450$$

$$59x = 1770$$

$$x = 30$$

Let,

The weight of silver sold in shop A = $22x = 22 * 30$ = 660 grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $660 + 240$ = 900 grams

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The weight of silver sold in shop B = $19x = 19 \times 30 = 570$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $570 + 690 = 1260$ grams

So,

The percentage of the total weight of silver in shop C = 18%

The total weight of silver in shop C = $(900/20) \times 18 = 810$ grams

The weight of silver sold in shop C = $(660/22) \times 21 = 630$ grams

The weight of silver unsold in shop C = $810 - 630 = 180$ grams

The percentage of the total weight of silver in shop D = 34%

The total weight of silver in shop D = $(900/20) \times 34 = 1530$ grams

The weight of silver sold in shop D = $(660/22) \times 38 = 1140$ grams

The weight of silver unsold in shop D = $1530 - 1140 = 390$ grams

Let,

The total weight of gold in shop D is 74 grams more than that of the total weight of silver in shop C.

The total weight of gold in shop D = $810 + 74 = 884$ grams

Then,

The total weight of gold in shop A = $(884/34) \times 20 = 520$ grams

The total weight of gold in shop B = $(884/34) \times 28 = 728$ grams

The total weight of gold in shop C = $(884/34) \times 18 = 468$ grams

Let table,

Shop	Gold (in grams)			Silver (in grams)		
	Total	Sold	Unsold	Total	Sold	Unsold
A	520	-	-	900	660	240
B	728	-	-	1260	570	690
C	468	-	-	810	630	180
D	884	-	-	1530	1140	390

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Answer: C

The sum of the weight of gold unsold in shops B and D together = $(728 + 884) - 1158 = 454$ grams

The weight of gold unsold in shop B = $(454/227) \times 91 = 182$ grams

The weight of gold unsold in shop D = $454 - 182 = 272$ grams

The difference between the weight of gold unsold in shop B to that in shop D = $272 - 182 = 90$

The difference between the weight of silver unsold in shop B to that in shop D = $690 - 390 = 300$

Thus, ratio = $90:300 = 3:10$

Hence, option C is correct.

8. Questions

From pie chart 2,

The total weight of silver sold combined in all four shops together = $100x$

The weight of silver sold in shop A = 22% of $100x = 22x$ grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $22x + 240$ grams

The weight of silver sold in shop B = 19% of $100x = 19x$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $19x + 690$ grams

The percentage of the total weight of silver in each shop is same as that of gold in each shop respectively.

The percentage of the total weight of gold in shop A = 20%

The percentage of the total weight of gold in shop B = 28%

Let,

$$(22x + 240)/(19x + 690) = 20/28$$

$$(22x + 240) * 7 = (19x + 690) * 5$$

$$154x + 1680 = 95x + 3450$$

$$59x = 1770$$

$$x = 30$$

Let,

The weight of silver sold in shop A = $22x = 22 * 30 = 660$ grams

The weight of silver unsold in shop A = 240 grams

The total weight of silver in shop A = $660 + 240 = 900$ grams

The weight of silver sold in shop B = $19x = 19 * 30 = 570$ grams

The weight of silver unsold in shop B = 690 grams

The total weight of silver in shop B = $570 + 690 = 1260$ grams

So,

The percentage of the total weight of silver in shop C = 18%

The total weight of silver in shop C = $(900/20) * 18 = 810$ grams

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The weight of silver sold in shop C = $(660/22) * 21 = 630$ grams

The weight of silver unsold in shop C = $810 - 630 = 180$ grams

The percentage of the total weight of silver in shop D = 34%

The total weight of silver in shop D = $(900/20) * 34 = 1530$ grams

The weight of silver sold in shop D = $(660/22) * 38 = 1140$ grams

The weight of silver unsold in shop D = $1530 - 1140 = 390$ grams

Let,

The total weight of gold in shop D is 74 grams more than that of the total weight of silver in shop C.

The total weight of gold in shop D = $810 + 74 = 884$ grams

Then,

The total weight of gold in shop A = $(884/34) * 20 = 520$ grams

The total weight of gold in shop B = $(884/34) * 28 = 728$ grams

The total weight of gold in shop C = $(884/34) * 18 = 468$ grams

Let table,

Shop	Gold (in grams)			Silver (in grams)		
	Total	Sold	Unsold	Total	Sold	Unsold
A	520	-	-	900	660	240
B	728	-	-	1260	570	690
C	468	-	-	810	630	180
D	884	-	-	1530	1140	390

Answer: B

The sum of the weight of silver sold in all four shops together = $660 + 570 + 630 + 1140 = 3000$ grams

The sum of the weight of gold sold in all four shops together = 59.4% of 3000 = 1782 grams

The sum of the weight of gold unsold in all four shops together = $(520 + 728 + 468 + 884) - 1782 = 818$ grams

The sum of the weight of silver in all four shops together = $900 + 1260 + 810 + 1530 = 4500$ grams

Thus, percentage = $(818/4500) * 100 = (818/45)\%$

Hence, option B is correct.

9. Questions

$$X + U = 170 \text{ ----- (1)}$$

$$U - X = 10 \text{ ---->(2)}$$

Solve the equations (1) and (2)

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$$2U = 180$$

$$U = 90$$

And

$$X + 90 = 170$$

$$X = 80$$

Attempt 1

The total number of students who appeared for the exam = 280

The number of students who passed in the exam = 80% of 280 = 224

The number of students who failed the exam = 280 – 224 = 56

Attempt 2

The total number of students who appeared for the exam = 274 + 56 = 330

The number of students who failed the exam = 89 – 56 = 33

The number of students who passed in the exam = 330 – 33 = 297

Attempt 3

The total number of students who appeared for the exam = 182 + 33 = 215

The number of students who passed in the exam = 80% of 215 = 172

The number of students who failed the exam = 215 – 172 = 43

Attempt 4

The number of students who failed the exam = 156 – 43 – 33 – 56 = 24

The number of students who passed in the exam = (24/8) * 92 = 276

The total number of students who appeared for the exam = (276/2) * 100 = 300

The number of new students who appeared for the exam = 300 – 43 = 257

Let table,

Attempts	The number of new students who appeared for the exam	The total number of students who appeared for the exam	The number of students who passed the exam	The number of students who failed the exam
1	280	280	224	56
2	274	330	297	33
3	182	215	172	43
4	257	300	276	24

Answer: D

Let,

The number of male students who failed the exam in attempt 2 = $(33/11) * 6 = 18$

The number of female students who failed the exam in attempt 2 = $(33/11) * 5 = 15$

The number of male students who passed the exam in attempt 2 = $2X = 2 * 80 = 160$

The number of female students who passed the exam in attempt 2 = $297 - 160 = 137$

The number of female students who appeared the exam in attempt 2 = 137

+15 = 152

So,

$2X - 8 = 160 - 8 = 152$

Hence, option D is correct.

10. Questions

$$X + U = 170 \text{ ---- (1)}$$

$$U - X = 10 \text{ ---->(2)}$$

Solve the equations (1) and (2)

$$2U = 180$$

$$U = 90$$

And

$$X + 90 = 170$$

$$X = 80$$

Attempt 1

The total number of students who appeared for the exam = 280

The number of students who passed in the exam = 80% of 280 = 224

The number of students who failed the exam = $280 - 224 = 56$

Attempt 2

The total number of students who appeared for the exam = $274 + 56 = 330$

The number of students who failed the exam = $89 - 56 = 33$

The number of students who passed in the exam = $330 - 33 = 297$

Attempt 3

The total number of students who appeared for the exam = $182 + 33 = 215$

The number of students who passed in the exam = 80% of 215 = 172

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The number of students who failed the exam = $215 - 172 = 43$

Attempt 4

The number of students who failed the exam = $156 - 43 - 33 - 56 = 24$

The number of students who passed in the exam = $(24/8) * 92 = 276$

The total number of students who appeared for the exam = $(276/2) * 100 = 300$

The number of new students who appeared for the exam = $300 - 43 = 257$

Let table,

Attempts	The number of new students who appeared for the exam	The total number of students who appeared for the exam	The number of students who passed the exam	The number of students who failed the exam
1	280	280	224	56
2	274	330	297	33
3	182	215	172	43
4	257	300	276	24

Answer: C

The number of students who passed the exam from the number of new students who appeared for the exam = 85.71% of $182 = (6/7) * 182 = 156$

The remaining students who passed were those who had failed in the previous attempt = $172 - 156 = 16$

The number of students who failed the exam from the number of new students who appeared for the exam = $182 - 156 = 26$

The number of students who failed again from the previous attempt = $33 - 16 = 17$

Thus, difference = $26 - 17 = 9$

Hence, option C is correct.

11. Questions

$$X + U = 170 \text{ ----- (1)}$$

$$U - X = 10 \text{ ---->(2)}$$

Solve the equations (1) and (2)

$$2U = 180$$

$$U = 90$$

And

$$X + 90 = 170$$

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$$X = 80$$

Attempt 1

The total number of students who appeared for the exam = 280

The number of students who passed in the exam = 80% of 280 = 224

The number of students who failed the exam = 280 – 224 = 56

Attempt 2

The total number of students who appeared for the exam = 274 + 56 = 330

The number of students who failed the exam = 89 – 56 = 33

The number of students who passed in the exam = 330 – 33 = 297

Attempt 3

The total number of students who appeared for the exam = 182 + 33 = 215

The number of students who passed in the exam = 80% of 215 = 172

The number of students who failed the exam = 215 – 172 = 43

Attempt 4

The number of students who failed the exam = 156 – 43 – 33 – 56 = 24

The number of students who passed in the exam = (24/8) * 92 = 276

The total number of students who appeared for the exam = (276/2) * 100 = 300

The number of new students who appeared for the exam = 300 – 43 = 257

Let table,

Attempts	The number of new students who appeared for the exam	The total number of students who appeared for the exam	The number of students who passed the exam	The number of students who failed the exam
1	280	280	224	56
2	274	330	297	33
3	182	215	172	43
4	257	300	276	24

Answer: E

The difference between the number of students who passed the exam in attempts 2 and 4 = 297 – 276 = 21

Thus, percentage = (21/300) * 100 = 7%

Hence, option E is correct.

12. Questions

$$X + U = 170 \text{ ---- (1)}$$

$$U - X = 10 \text{ ---->(2)}$$

Solve the equations (1) and (2)

$$2U = 180$$

$$U = 90$$

And

$$X + 90 = 170$$

$$X = 80$$

Attempt 1

The total number of students who appeared for the exam = 280

The number of students who passed in the exam = 80% of 280 = 224

The number of students who failed the exam = 280 – 224 = 56

Attempt 2

The total number of students who appeared for the exam = 274 + 56 = 330

The number of students who failed the exam = 89 – 56 = 33

The number of students who passed in the exam = 330 – 33 = 297

Attempt 3

The total number of students who appeared for the exam = 182 + 33 = 215

The number of students who passed in the exam = 80% of 215 = 172

The number of students who failed the exam = 215 – 172 = 43

Attempt 4

The number of students who failed the exam = 156 – 43 – 33 – 56 = 24

The number of students who passed in the exam = $(24/8) * 92 = 276$

The total number of students who appeared for the exam = $(276/2) * 100 = 300$

The number of new students who appeared for the exam = 300 – 43 = 257

Let table,

Attempts	The number of new students who appeared for the exam	The total number of students who appeared for the exam	The number of students who passed the exam	The number of students who failed the exam
1	280	280	224	56
2	274	330	297	33
3	182	215	172	43
4	257	300	276	24

Answer: A

The number of new students who appeared for the exam in attempt 5 = 140% of 90 = 126

The total number of new students who appeared for the exam in attempt 5 = 126 + 24 = 150

The number of students who passed the exam in attempt 5 = [40 + 10]% of 276 = 138

The number of students who failed the exam in attempt 5 = 150 – 138 = 12

Hence, option A is correct.

13. Questions

Answer: D

Compound Interest = $P * [1+(R/100)]^n - P$

$$(X - 10000)/(X + 2000) = 5/8$$

$$8X - 80000 = 5X + 10000$$

$$3X = 90000$$

$$X = 30000$$

The invested amount by A = $X - 10000 = 30000 - 10000 = \text{Rs. } 20000$

Interest rate of A = R%

In half yearly, let assume $R = 2r$

So, the interest rate = $2r/2 = r$

Let

$$20000 * [(100 + r)/100]^2 - 20000 = 2050$$

$$20000 * (100 + r)^2/10000 = 22050$$

$$10000 + r^2 + 200r = 11025$$

$$r^2 + 200r - 1025 = 0$$

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$$r^2 + 205r - 5r - 1025 = 0$$

$$r(r + 205) - 5(r + 205) = 0$$

$$(r - 5)(r + 205) = 0$$

$$r = 5 \text{ (positive number)}$$

$$R = 2r = 10$$

The invested amount by B = $X + 2000 = 30000 + 2000 = \text{Rs. } 32000$

Interest rate of B = $1.5R\% = 15\%$

The interest earned by B = $32000 * (115/100)^2 - 32000$

$$= 42320 - 32000$$

$$= \text{Rs. } 10320$$

Thus, difference = $10320 - 2050 = \text{Rs. } 8270$

Hence, option D is correct.

14. Questions

Answer: A

a' is a perfect cube in between $1000 \geq a < 1300$

So, only it matches

$$a = 1000$$

The cost price of the article = Rs. 1000

The selling price of the article = $1.8 * 1000 = \text{Rs. } 1800$

The marked price of the article = $200\% \text{ of } 1000 = \text{Rs. } 2000$

Let,

The new cost price of the article = $1000 + 100 = \text{Rs. } 1100$

The new selling price of the article = $1800 - 200 = \text{Rs. } 1600$

The new profit of the article = $1600 - 1100 = \text{Rs. } 500$

The new discount of the article = $2000 - 1600 = \text{Rs. } 400$

Thus, difference = $500 - 400 = \text{Rs. } 100$

Hence, option A is correct.

15. Questions

Answer: B

Let assume the total work = P

The efficiency of (A + B) = $P/(P/15) = 15$

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The efficiency of $(C + B) = P/(P/8) = 8$

The efficiency of $A = P/(P/10) = 10$

The efficiency of $B = 15 - 10 = 5$

The efficiency of $C = 8 - 5 = 3$

Then,

The time taken by C only complete the work in 50 days

$$P/3 = 50$$

$$P = 150$$

So,

$$(P/10)^2 - 1.2P$$

$$= (150/10)^2 - 1.2 * 150$$

$$= 45$$

$$P - 105 = 150 - 105 = 45$$

Hence, option B is correct.

16. Questions

Answer: C

Let assume the radius and height of the cone is 'r' cm and 'h' cm respectively

$$r + h = 49 \text{ ----(1)}$$

Let,

$$\text{The volume of the cone} = (4/3) * \pi * R^3$$

$$(4/3) * \pi * R^3 = 7776\pi$$

$$R^3 = 5832$$

$$R = 18 \text{ cm}$$

$$\text{The height of the cone} = 18 + 10 = 28 \text{ cm}$$

From (1)

$$r = 49 - 28$$

$$r = 21 \text{ cm}$$

$$\text{The radius of the cone} = 21 \text{ cm}$$

$$\text{The slant height of the cone} = l = \sqrt{(21^2 + 28^2)} = \sqrt{1225} = 35 \text{ cm}$$

Then

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The lateral surface area of the cone = πrl

$$= 22/7 * 21 * 35$$

$$= 2310 \text{ cm}^2$$

Hence, option C is correct.

17. Questions

Answer: C

The speed of train P = 10s m/sec

The speed of train Q = 70% of 10s = 7s m/sec

Let assume the length of trains P and Q is P m, and Q m, respectively.

The time taken by train P to cross the pole = $P/10s$

The time taken by train Q to cross the pole = $Q/7s$

Let,

$$(Q/7s)/(P/10s) = 2/1$$

$$(10Q)/(7P) = 2/1$$

$$Q/P = 7/5$$

$$P: Q = 5: 7 = 5x: 7x$$

Quantity I:

If the difference between the lengths of both trains P and Q is 192 m.

$$7x - 5x = 192$$

$$2x = 192$$

$$x = 96$$

The length of train P = $5x = 5 * 96 = 480 \text{ m}$

Quantity II:

The length of train P = $5x$

The time taken by train P to cross the 240 m platform is 1.5 times the time taken by train P to cross the pole.

$$(5x + 240)/10s = 1.5 * (5x/10s)$$

$$5x + 240 = 7.5x$$

$$2.5x = 240$$

$$x = 96$$

The length of train Q = $7 * 96 = 672 \text{ m}$

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The length of the bridge = $672 - 172 = 500$ m

Thus, Quantity I < Quantity II.

Hence, option C is correct.

18. Questions

Answer: C

Equation,

$$4p^2 + 19p + 56 * 24 = 2p^2 + 66p + 1159$$

$$2p^2 - 47p + 1344 - 1159 = 0$$

$$2p^2 - 47p + 185 = 0$$

$$2p^2 - 10p - 37p + 185 = 0$$

$$2p(p - 5) + 37(p - 5) = 0$$

$$(p - 5)(2p + 37) = 0$$

$$P = 5 \text{ and } 37/2$$

$$y = \text{smaller root} = 5$$

and,

$$5/(97 * 8 - 257 * 3) = 27/(z + 12)$$

$$5/(776 - 771) = 27/(z + 12)$$

$$5/5 = 27/(z + 12)$$

$$z + 12 = 27$$

$$z = 15$$

Quantity I:

$$100y = 100 * 5 = 500$$

Quantity II:

If the value of 'x' is 80% of the value of 'z'.

$$x = 80\% \text{ of } 15$$

$$x = 12$$

The value of ' $16.7x + 20z$ '.

$$= 16.7 * 12 + 20 * 15$$

$$= 200.4 + 300$$

$$= 500.4$$

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Thus, Quantity I < Quantity II.

Hence, option C is correct.

19. Questions

Answer: A

The present age of Y = x years

The present age of P = 3x years

The present age of A = 3x - 6 years

At the time of Y's birth

Age of P = 3x - x = 2x years

Age of A = 3x - 6 - x = 2x - 6 years

$$(2x - 6)/(2x) = 80/100$$

$$(x - 3)/x = 4/5$$

$$5x - 15 = 4x$$

$$x = 15$$

Then,

The present age of Y = x = 15 years

The present age of P = 3x = 45 years

The present age of A = 3x - 6 = 45 - 6 = 39 years

Quantity I:

The average age of the present age of P and the present age of A = $(45 + 39)/2 = 84/2 = 42$ years.

Quantity II:

If the ratio of the present age of S to the present age of Y is 13:3, respectively.

The present age of S = $(15/3) * 13 = 65$ years

The age of S 24 years ago = $65 - 24 = 41$ years.

Thus, Quantity I > Quantity II.

Hence, option A is correct.

20. Questions

Answer: E

Profit ratio of Prabhu and Banu

$$= [X * 5 + 120\% \text{ of } X * 7]: [Y * 6 + 75\% \text{ of } Y * 6]$$

$$= [5X + 8.4X]: [6Y + 4.5Y]$$

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$$= 13.4X: 10.5Y$$

The profit share of Prabhu = Rs. 8040

The profit share of Banu = $15600 - 8040 = \text{Rs. } 7560$

Let

$$13.4X/10.5Y = 8040/7560$$

$$X/Y = 600/720$$

$$X/Y = 5/6$$

$$X: Y = 5: 6$$

Quantity I:

The sum of the values of 'X' and 'Y' = Rs. 11000

The value of X = $(11000/11) * 5 = \text{Rs. } 5000$

Quantity II:

$$SI = PNR/1000$$

$$Y * 3 * 8/100 = 1440$$

$$Y = 6000$$

$$\text{Then, } X = 6000 * (5/6) = 5000$$

Thus, Quantity I = Quantity II

Hence, option E is correct.

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